

THERMABRICK · REQUIREMENTS

ThermaBrick requirements

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/thermabrick		
Source	docs/03-requirements.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.10	2026-10-01	Amish Chadha	Design made constructable (TBK-DDR-003). R15 at 415 kg; R20 full-scale estimate \$3,854 reported against the value-engineering target; R13 and R17 notes for the outlet trim ring and the screwed jacket cap
0.9	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.8	2026-09-24	Amish Chadha	Review pass: R20 shows \$727.70 as the BOM total, proposed and awaiting Amish; the accepted budget remains \$635.70
0.7	2026-09-24	Amish Chadha	R20 test article budget updated to \$727.70 after the build procedure's additions
0.6	2026-09-24	Amish Chadha	R20 records the decision to accept the test article at \$635.70
0.5	2026-09-24	Amish Chadha	R20 test article estimate updated to \$635.70 after the controller enclosure fit check (TBK-DWG-005)
0.4	2026-09-24	Amish Chadha	

R20 test article estimate updated to \$599.70 after adding the pull-down resistors

0.3	2026-09-24	Amish Chadha	Updated R3, R4 and R5 status for standby loss (TBK-CAL-001 v0.2), so R3 is now not met; R20 notes the \$600 reduced-scale test article
0.2	2026-09-24	Amish Chadha	Defined 20 requirements with targets, verification methods and v0.2 compliance status traced to TBK-CAL-001
0.1	2026-09-24	Amish Chadha	Initial scaffold

ThermaBrick requirements

The v0.2 design meets 16 of the 20 requirements below, by analysis, inspection or design. Three depend on firmware or on testing that has not yet been done (R12, R18 and R19). One is not met: charge time (R3) misses by 1.6 h once standby loss is counted. Amish accepted the recommendation to choose how to close R3 only after the test article has measured sand conductivity, so R3 stays open while TRL 4 is on hold (TBK-DDR-002). R20 is restated to match the budget decisions: the prototype budget covers the reduced-scale test article, set at \$727.70 by Amish on 2026-09-25, and the full-scale estimate is reported against it. The targets of R1 to R20 were accepted by Amish on 2026-09-25 (TBK-DDR-002). Every requirement will be confirmed by test on the first build.

Verification methods are analysis (A), inspection (I), demonstration (D) and test (T). Status reflects the v0.2 analysis in TBK-CAL-001 and the BOM. "Met (A)" means shown by analysis and not yet tested.

■ 1. Performance

ID	Requirement	Target	Verification	v0.2 status
R1	Stored heat between 150 °C and 450 °C energy-weighted mean sand temperature	18 kWh(th) or more	A: TBK-CAL-001 section 2. T: calorimetric discharge (air flow and temperature rise)	Met (A): 18.3 kWh(th)
R2	Charge power drawn only from PV surplus, following export in real time	0 to 3.0 kW in steps of 25 W or finer; grid import from charging 50 Wh per day or less	D: log export meter and heater power through three sunny days	Met by design
R3	Charge acceptance from empty with full surplus available	11 kWh or more in 4 h; full in 8 h or less	A: TBK-CAL-001 section 3. T: charge from 150 °C at 3.0 kW	Not met (A): 10.9 kWh in 4 h, full in 9.6 h. The fix is chosen after the test article measures sand conductivity (TBK-DDR-002); on hold with TRL 4
R4	Rated heat output	1.0 kW held down to 170 °C sand mean or lower	A: TBK-CAL-001 section 4. T: discharge at 1.0 kW demand	Met (A): held 14.3 h, to 154 °C
R5	Boost heat output	1.5 kW for 8 h or more from full	A: TBK-CAL-001 section 4. T	Met (A): 8.1 h
R6	Supply air temperature at the register	55 °C or below at all times	T: log supply air through charge and discharge	Met by design (50 °C setpoint)
R7	Standby loss	500 W or less at full charge; 9 kWh or less over 24 h idle from full	A: TBK-CAL-001 section 5. T: 24 h cool-down from full	Met (A): 459 W; 8.7 kWh

■ 2. Safety

ID	Requirement	Target	Verification	v0.2 status
R8	Peak temperatures	Sand and well wall 550 °C or below; heater sheath 700 °C or below; drum wall 500 °C or below	A: TBK-CAL-001 section 3. T: well-wall and sand thermocouples, plus a temporary drum-wall thermocouple beside a ring B well	Met (A): 550 °C, 634 °C and about 480 °C
R9	Accessible surface temperature at full charge, 20 °C room	45 °C or below on the jacket; hot outlet duct guarded	A: TBK-CAL-001 section 5. T: surface survey with a contact probe	Met (A): 26 °C side
R10	Independent over-temperature protection	Hardware limit, separate from the ESP32, opens a 2-pole contactor at 600 °C well-wall temperature and latches until manually reset	I: wiring against the schematic. T: trip test with a heated thermocouple	Met by design
R11	Electrical supply and protection	Dedicated 240 V, 20 A circuit with 2-pole GFCI protection; continuous current 16 A or less; supplementary fuses per heater group	A: TBK-CAL-001 section 6. I	Met (A): 12.5 A
R12	Fail-safe behavior	On controller reset, loss of Wi-Fi, a sensor fault or a fan fault: heaters off, inlet damper closed	D: fault injection for each case	Firmware to be written
R13	Materials in hot zones	No zinc, paint, liner or plastic above 200 °C; AES (body-soluble) fiber in place of refractory ceramic fiber	I: against the BOM	Met by design. The hot outlet passes the galvanized jacket cap in a 180 mm hole packed with AES and covered by a black steel trim ring, so no zinc sits within 39 mm of it (TBK-DDR-003)

■ 3. Physical and installation

ID	Requirement	Target	Verification	v0.2 status
R14	Envelope	1,250 mm diameter or less; 1,500 mm high or less, including ducts	I: CAD and as-built	Met: 1,209 mm by 1,462 mm
R15	Siting and floor load	Indoors, inside the heated space, on a concrete slab; 450 kg or less in service	A: TBK-CAL-001 section 7	Met (A): 415 kg (TBK-CAL-001 v0.3)
R16	Buildability	No welding, machining or pressure parts; hand tools, a pipe threader and pop rivets only	I: build log	Met by design
R17	Serviceability	Any heater replaceable from the top without removing sand	D: replace one heater	Met by design: heaters sit in wells; the jacket cap is screwed, not riveted, so it and the top insulation lift off to reach them (TBK-DDR-003)

■ 4. Controls, openness and cost

ID	Requirement	Target	Verification	v0.2 status
R18	Monitoring	Log all temperatures, heater power and export at 10 s intervals; report state of charge within 10 % of the calorimetric value	T: compare against R1 test	Firmware to be written
R19	Durability	1,000 charge cycles without drum growth over 1 % in circumference or loss of heater function	T: measure drum circumference every 25 cycles	Open: sand ratcheting risk, see TBK-PRC-001 section 8
R20	Prototype cost	Reduced-scale test article (TBK-PRC-002, archived): \$727.70 USD or less in parts, decided by Amish on 2026-09-25 (TBK-DDR-002). Full-scale parts cost estimated and reported; its budget is revisited with the test article's results	I: against the BOMs	Met (I): test article BOM \$727.70 (archived, on hold with TRL 4). Full-scale estimate of the constructable design \$3,854 reported, USD 3,126 over the USD 728 value-engineering target; revisit on hold with TRL 4

■ 5. Traceability

The requirements trace to TBK-PRB-001 as follows: R1 to R7 to section 4 (success measures); R8 to R13 to section 3 (safety constraint); R14 to R17 to section 3 (buildability and siting); R18 to section 4 (reproducibility); R20 to section 3 (budget). R19 comes from the ratcheting risk identified in the design precis.